

KEEP THE ANGLE: A GEOMETRY-PRESERVING BASIS IN SPECTRAL EMBEDDINGS

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Abstract. The commute-time (resistance) embedding built from the low eigenvectors of a graph Laplacian is a workhorse of spectral geometry, yet its distances degenerate on large graphs (von Luxburg, Radl, and Hein). We show the geometry does not vanish: it is largely carried by the *angular* coordinate. Writing each node’s low-mode embedding in polar form—magnitude times direction—the direction preserves graph geodesics (Spearman $\rho \approx 0.93$ on a reference torus), while the magnitude, and an equal-size random-mode representation, preserve almost none. Keeping only the direction is the row-normalization of Ng–Jordan–Weiss spectral clustering applied to the commute-weighted embedding; its existing theory (Schiebinger, Wainwright, and Yu) concerns cluster separation, and we supply a metric-geometric account. An exact identity locates the mechanism in *tangential noncollapse*—the embedding velocity must not turn purely radial—rather than in radius constancy; from it we prove a deterministic angular-transfer theorem, a conditional graph-to-manifold result (carried by the random-walk eigenvectors and the sup-norm eigenvector rates of Calder, García Trillos, and Lewicka), and an *unconditional* bi-Lipschitz statement on the flat torus. The radial claim is stated at three levels: proved for the full embedding at intrinsic dimension $d \geq 3$, empirical for the truncated radius the algorithm computes, and borderline at $d = 2$. Uniformity of the metric constants in the mode count fails for the commute filter, but the *rank* form is uniform below dimension four: for $d \leq 3$ the angular ranking converges to a fixed Green-kernel ranking, exact on the circle and on two-point homogeneous spaces. The extended empirical program—substrate robustness across graph families, a manifold diagnostic, and the fidelity–dimension trend—is the companion Paper I.b.

Key words. spectral embedding, commute-time (resistance) distance, row normalization, spectral clustering, geodesic distance, intrinsic dimension, Laplacian eigenmaps

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1. Introduction. Spectral embeddings—Laplacian eigenmaps [3], diffusion and commute-time maps [9]—represent a graph by the low eigenvectors of its Laplacian and underpin much of manifold learning and spectral clustering. They carry a well-known pathology: on large geometric graphs the commute (resistance) distance degenerates to a function of local degrees alone, losing all global geometry [26]. Yet the most successful spectral-clustering algorithm [17] *row-normalizes* the embedding—projects each node to the unit sphere—and works well. The existing theory of that normalization concerns cluster geometry: it explains why row-normalized embeddings of well-separated clusters concentrate in nearly orthogonal directions on the sphere [23, 25]. It does not address metric geometry. Closer to our object, unit-hypersphere normalization has been applied *directly* to the commute-time embedding for 3D shape registration across differing samplings [24]; there it is a sampling-robustness heuristic, and the geodesic content of the resulting angular distances is not analyzed. **This paper asks what row-normalization recovers of the graph’s geodesic structure, and why.**

Our answer is a clean decomposition. Write each node’s low-mode embedding in polar form, magnitude $\times$ direction. The *magnitude* is the truncated, normalized counterpart of the radial coordinate that von Luxburg’s theorem kills for the *full, unnormalized* commute embedding at $d \geq 3$ (Corollary 3.1); empirically it tracks local degree (§5.2). The *direction* carries the graph’s geodesic geometry (after the $\alpha=1$ density normalization where sampling is strongly non-uniform, Paper I.b). Row-

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normalization keeps the direction and discards the magnitude—so beyond its cluster-separation role it is also the geodesically correct projection, and its fidelity is flat in both mode-count and graph size precisely where the raw embedding decays.

The underlying operation—*separate scale from direction, then keep the part that carries the task geometry*—recurs beyond clustering: polar-coordinate KV-cache quantization splits key vectors the same way [14], evidence for the *decomposition*, not for always discarding the radius. Our hypotheses fix the boundary: discarding the magnitude is the right quotient precisely when the radial variable is nuisance for the metric preserved—the tangential-noncollapse condition (A2) of Theorem 3.8—which for graph geodesics holds, the radius being degree/density noise (Corollary 3.1). On the reference torus the direction recovers geodesics where the magnitude and a random-mode representation do not (§5); the companion Paper I.b [1] establishes the basis’s substrate-robustness and its failure modes across graph families.

Contributions. The first is the **Keep-the-Angle Principle** (§3), which comprises: a Radial-Degeneracy corollary for the full embedding (Corollary 3.1, from [26]), with truncation controlled by $1/\lambda_{m+1}$ (Proposition 3.3); an exact angular-distortion identity raised to a deterministic angular-transfer theorem for arbitrary filtered embeddings (Theorem 3.8, Corollary 3.9), from which Angular Preservation follows conditionally on the graph (Theorem 3.10, section SM1) and unconditionally on the flat torus and round sphere (Corollaries 3.12, 3.13); a distortion-to-rank bridge (Proposition 3.14); and the filter dichotomy, with a uniform lower angular bound and a Green-kernel rank limit for $d \leq 3$ (Theorem SM2.7). The second is a confirming measurement on the torus—controls (§5), theorem verification (§5.2), and a weighting ablation (§5.3); the extended empirical program is the companion Paper I.b [1]. We do not claim to derive physics: the core result is a basis-level fact about spectral embeddings.

TABLE 1.1
Status of the paper’s claims at a glance.

Claim	Status	Where
Radial degeneracy (full commute)	proved (vL–R–H + bdd. deg.)	Cor. 3.1
Truncated normalized radius $\sim$ degree	empirical	§5.2
Truncation controlled by $1/\lambda_{m+1}$	proved	Prop. 3.3
Exact angular-distortion identity	proved	Prop. 3.7
Deterministic transfer (A1)–(A3)	proved	Thm. 3.8
Graph transfer at fixed m	proved, cond. (H1)–(H2)	Thm. 3.10
Global form	cond. (H3)	Cor. 3.11
Flat torus, $m = 4$	proved, uncond.	Cor. 3.12
Round sphere, $m = d + 1$	proved, uncond.	Cor. 3.13
Distortion $\Rightarrow$ rank	proved	Prop. 3.14
Heat filter, uniform in m	proved (continuum)	Thm. 3.15
Commute <i>upper</i> , fixed-scale unif.	refuted (exact $\mathbb{S}^1$)	Prop. 3.16
Commute <i>lower</i> bound, unif. in m (tori)	proved	Prop. SM2.3
Rank limit = Green-kernel ranking ($d \leq 3$)	proved	Thm. SM2.7
Filter phase diagram, critical $d = 4$	proved	Prop. SM2.5
Plain-filter rank collapse on $\mathbb{S}^1$	proved	Thm. 3.18
Angular preservation (rank form)	conjecture; empirical	Conj. 3.4
Unnormalized differential noncollapse	proved (monotone)	Prop. 3.5

2. Background. Spectral geometry. Laplacian eigenmaps [3] and diffusion maps [9] embed a graph via low eigenvectors; the commute-time embedding scales mode k by $1/\sqrt{\lambda_k}$. **Commutetime degeneracy.** von Luxburg et al. [26] proved

the commute distance of a large geometric graph degenerates, $R(i, j) \rightarrow 1/k_i + 1/k_j$ (k_i the degree of node i); we read this backwards as the collapse of the *radial* coordinate to local degree. They also propose an *amplified* commute distance as a correction, the natural competitor to ours; rather than repair the radius we discard it and keep the angle. **Row-normalization.** Ng–Jordan–Weiss [17] row-normalize the eigen-embedding—the angular projection we study. Schiebinger, Wainwright and Yu [23] give the sharpest existing account: under a mixture model, row-normalized embeddings concentrate clusters in orthogonal directions, explaining k -means’ post-normalization success; consistency of spectral clustering is treated in [25]. Both analyses are about cluster separation on the sphere. Our question—whether the normalized embedding preserves *geodesic distances* on a manifold-like graph—is complementary, and to our knowledge open. The same normalization removes degree as a nuisance in degree-corrected block models—SCORE [15] cancels it through eigenvector ratios, and spherical spectral clustering under the DCBM [20] projects to the sphere—but those are community-recovery guarantees, not geodesic ones. The embedding Ψ itself is Rustamov’s Global Point Signature [21] from shape analysis, and the magnitude we discard is used in its own right for cluster and anomaly detection [8]; we keep the direction and give it a metric reading. **Hypergraph-rewriting graph families.** Hypergraph rewriting [27, 12] generates large graphs from a local rule; we use several such rules as one of five substrate families, treated purely as graph generators (any physical interpretation is deferred to §6). **Hyperbolic embedding.** Trees embed in $\mathbb{H}^2$ at low distortion [13, 22, 18]; we test and reject a hyperbolic reframe in Paper I.b.

3. The Keep-the-Angle Principle. Let G be connected on n nodes with symmetric-normalized Laplacian $L = I - D^{-1/2}AD^{-1/2}$, eigenpairs (λ_k, v_k) , $0 = \lambda_0 < \lambda_1 \leq \dots$. The low-mode spectral embedding using the lowest m nontrivial modes places node i at $\Psi_i = (v_1(i)/\sqrt{\lambda_1}, \dots, v_m(i)/\sqrt{\lambda_m})$, the eigenvectors unit-normalized ($\sum_i v_k(i)^2 = 1$, i.e. $VV^\top = I$; the empirical convention $\frac{1}{n} \sum_i v_k(i)^2 = 1$ rescales the numerical truncation bounds by n but leaves all angular statements, which are scale-invariant, unchanged). Write $\Psi_i = r_i \hat{u}_i$ with radius $r_i = \|\Psi_i\|$ and angle $\hat{u}_i = \Psi_i/\|\Psi_i\|$. Throughout, “angular distance” means the Euclidean *chordal* distance between unit vectors, $\|\hat{u}_i - \hat{u}_j\| = 2 \sin(\theta_{ij}/2)$ with θ_{ij} the angle between $\hat{u}_i$ and $\hat{u}_j$; it is monotone in θ_{ij} , so all rank (Spearman) statements are unchanged by the chordal-vs-arc choice, and the figures use the chordal form.¹

Intrinsic dimension and degree notation. Throughout, d is the intrinsic dimension of the substrate, and k_i denotes the degree of node i (we reserve d for dimension to avoid the collision d vs. d_i): the manifold dimension *by construction* for the random-geometric benchmarks (tori, king-lattices), and for emergent graphs the value at which the three independent estimators of §4 (ball-growth, spectral dimension, effective rank) mutually agree—reported with their inter-estimator spread, with only low-spread graphs treated as manifolds. Dimension regimes use this measured intrinsic dimension, with one convention enforced throughout: the radial-degeneracy *theorem* is gated at $d \geq 3$; $d = 2$ is the recurrent borderline, where the degeneracy appears with logarithmic corrections and is treated empirically (§5.2); and in (quasi-)1D the radius stays geodesic-informative, as predicted (Paper I.b). The angular *effect* is observed from quasi-1D upward.

¹We call Ψ “commute-style” rather than the commute-time embedding proper: the classical commute embedding uses the pseudoinverse of the *unnormalized* Laplacian and all $n - 1$ modes, so that $\|\Psi_i - \Psi_j\|^2$ equals the resistance $R(i, j)$ exactly. Our Ψ is the truncated, normalized analogue standard in spectral clustering; the distinction matters for Corollary 3.1 and is handled there.

COROLLARY 3.1 (Radial Degeneracy). *Let G_n be random geometric graphs of intrinsic dimension $d \geq 3$ satisfying the conditions of [26] (the $d = 2$ case is marginal: at the recurrent boundary the resistance carries logarithmic corrections, matching the borderline behaviour we measure in §5.2), and let Φ_i be the full commute-time embedding built from the unnormalized Laplacian, so that $\|\Phi_i\|^2 = L_{ii}^+$. Assume additionally that the degree ratio $\max_i k_i / \min_i k_i$ stays bounded (as it does with high probability in the graph models of [26]). As $n \rightarrow \infty$ the resistance distance degenerates uniformly and multiplicatively, $\sup_{i \neq j} |R(i, j)/(1/k_i + 1/k_j) - 1| \rightarrow 0$ [26], and this pairwise statement transfers to the diagonal: $L_{ii}^+ = (1/k_i)(1 + o(1))$ uniformly over nodes, so the radius satisfies $\|\Phi_i\| = (1/\sqrt{k_i})(1 + o(1))$. The radius of the full commute embedding is therefore asymptotically geometry-free—a local-density coordinate carrying no geodesic information.*

The diagonal transfer is not automatic: the pairwise limits determine only combinations $L_{ii}^+ + L_{jj}^+ - 2L_{ij}^+$, and a gauge ($\sum_j L_{ij}^+ = 0$) plus a trace identity, with the degree-ratio bound controlling error propagation, close the gap (Appendix A).

Remark 3.2 (Truncated, normalized radius). Our experiments use the truncated radius $r_i = \|\Psi_i\|$ (m lowest modes of L_{sym}), not $\|\Phi_i\|$. Transferring Corollary 3.1 to this setting needs (i) the normalized analogue of the degeneracy [26] and (ii) control of the truncation error. We settle (ii) unconditionally below and support (i) numerically; the residual gap is then the angular claim itself (Conjecture 3.4). Numerically the truncated radius tracks the local degree, $\rho(r_i, 1/\sqrt{k_i}) = 0.79\text{--}0.87$ across $m = 5\text{--}40$, while the full normalized radius is near-constant (coefficient of variation 0.09) and radius-only geodesic preservation is $\rho \leq 0.08$ (§5.2).

PROPOSITION 3.3 (Inverse-next-eigenvalue truncation bound). *Let $\Psi_i^\infty = (v_k(i)/\sqrt{\lambda_k})_{k=1}^{n-1}$ be the full symmetric-normalized commute-style embedding and Ψ_i^m its truncation to the lowest m nontrivial modes. Then for all nodes $i \neq j$,*

$$0 \leq \|\Psi_i^\infty\|^2 - \|\Psi_i^m\|^2 \leq \frac{1}{\lambda_{m+1}}, \quad 0 \leq \|\Psi_i^\infty - \Psi_j^\infty\|^2 - \|\Psi_i^m - \Psi_j^m\|^2 \leq \frac{2}{\lambda_{m+1}}.$$

Proof. Each left inequality is termwise: truncation deletes only the summands with $k > m$, of the form $v_k(i)^2/\lambda_k \geq 0$ or $(v_k(i) - v_k(j))^2/\lambda_k \geq 0$. For the right inequalities, $\lambda_k \geq \lambda_{m+1}$ for every $k > m$, so the deleted mass is at most $\lambda_{m+1}^{-1} \sum_{k>m} v_k(i)^2$ and $\lambda_{m+1}^{-1} \sum_{k>m} (v_k(i) - v_k(j))^2$ respectively. The eigenvectors $\{v_k\}_{k=0}^{n-1}$ are orthonormal, so $VV^\top = I$ gives $\sum_k v_k(i)^2 = 1$ and, for $i \neq j$, $\sum_k v_k(i)v_k(j) = 0$; hence $\sum_{k>m} v_k(i)^2 \leq 1$ and $\sum_{k>m} (v_k(i) - v_k(j))^2 \leq \sum_k (v_k(i) - v_k(j))^2 = \sum_k v_k(i)^2 + \sum_k v_k(j)^2 - 2\sum_k v_k(i)v_k(j) = 2$. These identity sums run over all $k = 0, \dots, n-1$, including the trivial mode $k = 0$ that Ψ omits; it sits only in the subtracted low-mode partial sums, and dropping its nonnegative terms can only tighten the tail bounds. $\square$

Two consequences. First, the error is controlled by the inverse spectral gap $1/\lambda_{m+1}$ alone. This is *not* an n -independent absolute bound: on a fixed-degree neighbourhood graph refined toward the continuum λ_{m+1} itself shrinks, so $1/\lambda_{m+1}$ grows. What it controls is the truncation error *relative to the embedding scale*, which is fixed by the same low eigenvalues; since the angular and rank statistics we report are invariant to global scale, that relative control is what transfers, and is why the continuum refinement of Paper I.b does not degrade. Second, the deleted content is high-frequency (large- λ); the geometry lives in the retained low modes (an equal-size random-mode basis preserves $\rho \approx 0$, §5). But this bound is a *sanity check*, *not the angular theorem*:

it controls the full-vs-truncated discrepancy at a fixed graph and mode cutoff, in a norm weighted by inverse eigenvalues; it does not show the deleted tail is irrelevant to angular *rank* geometry, and it gives no uniform-in- n angular stability statement. The empirical flatness in m (§5.2) remains *evidence for* Conjecture 3.4, not a consequence of Proposition 3.3. On the 2-torus ($n = 2000$) the bound holds with room to spare: the realized radius error is 3.5–17% of $1/\lambda_{m+1}$ across $m \in \{5, 10, 20, 40\}$, the full normalized radius is near-constant (coefficient of variation 0.09), and the truncated radius stays rank-correlated with $1/\sqrt{k_i}$. What remains open is not the radius but the angle: Proposition 3.3 bounds how far truncation moves the metric, not that the surviving *angular* coordinate equals the geodesic—that is Conjecture 3.4.

CONJECTURE 3.4 (Angular Preservation). *For ε -graph or k -NN samples of a compact Riemannian manifold in the density class of [11, 6], row-normalization $\hat{u}_i = \Psi_i / \|\Psi_i\|$ deletes the degenerate density factor of Corollary 3.1, and the pairwise angular distances remain rank-faithful to geodesic distance—Spearman $\geq \rho_0 > 0$ —uniformly over the admissible mode schedule: m multiplet-closed at a spectral gap with $m \geq m_0(M)$ (the immersion threshold of [2, 19]); sampled pairs above the resolution scale θ_n ; and, for growing $m = m(n)$, the spectral-window condition $\mathcal{E}_n(\mu_m)\gamma_m^{-1} \rightarrow 0$ of section SM1. Under strongly non-uniform sampling the claim is asserted for the Coifman–Lafon $\alpha=1$ angle (Paper I.b); empirically $\rho_0 \approx 0.9$.*

We reserve “rank-faithful” for the limit $\liminf_n \rho_S \geq \rho_0 > 0$ along admissible schedules $(n, m(n))$, with $\rho_0 = \rho_0(M)$ per manifold (the Green-rank coefficient of Theorem SM2.7 for $d \leq 3$); the diagnostic reading—using the small-world/dimension-spread screen of Paper I.b to decide whether a *given* graph lies in the class—is a separate corollary-in-use, kept out of the conjecture to avoid circularity. “Uniformly” means a lower bound over the admissible family, not over all integers m : too few modes need not embed M , a truncation inside a multiplet is basis-dependent, and m may not outrun the sample.

A stronger *metric* form—bi-Lipschitz to geodesic distance with constants uniform in m and n at fixed scale—is now known **false** for the commute weighting: high modes raise the local angular speed like $\sqrt{\Lambda}$ (Proposition 3.16, exact on S^1), so the metric strengthening must be either scale-dependent (uniform above the wavelength $\Lambda^{-1/2}$, open) or filtered (for the heat weighting it is a continuum theorem, Theorem 3.15). The rank clause is untouched: the empirical flatness in m is rank stability at a fixed mode budget, and our evidence is rank-based.

Metric evidence on the torus.. On the 2-torus we compare embedded distance to graph geodesic directly. After a single global scale the angular map has a robust empirical bi-Lipschitz constant $\kappa = 2.66 \pm 0.03$ (ratio of the 97.5th to 2.5th percentiles of $d_{\text{emb}}/d_{\text{geo}}$ over 19,900 pairs, five independent draws), against 3.15 ± 0.04 for the full commute embedding, 7.1 ± 0.3 for an equal-size random-mode basis, and 128 ± 4 for the magnitude; its leading four coordinates Procrustes-match the canonical flat-torus embedding in $\mathbb{R}^4$ at disparity 0.064 ± 0.016 (random modes: 0.985 ± 0.004); the map stays injective—nothing is folded together. This is metric distortion, not rank correlation—one substrate, but a well-understood one: the correct uniform metric target is scale-dependent for the commute filter (Proposition 3.16).

The uniform-in- m frontier.. The metric form of Conjecture 3.4 rests on one named hypothesis, which we isolate rather than assume away.

(H1)-uniform. *At mode-counts m taken at a spectral gap, the truncated eigenmap $F = (\varphi_k / \sqrt{\mu_k})_{k \leq m}$ is L_0 -bi-Lipschitz onto its image on geodesic r_0 -balls, with L_0 and r_0 independent of m .*

What is known is weaker: [5] and the quantitative truncated-eigenmap embeddings of [2, 19] give, for each fixed m past an immersion threshold, *some* constant $L_0(m)$, and as m grows the added eigenfunctions oscillate on wavelength $\mu_m^{-1/2}$, so holding L_0 fixed asks that no mode ever create a near-fold—the degeneration a symmetry or thin neck can force. Two endpoints bound the question: on the flat torus the *two-sided* bound holds at the first multiplet (Corollary 3.12, the Clifford embedding) and the *lower* half holds uniformly in m (Proposition SM2.3), and the property is *filter-specific*—a theorem for the heat weighting (Theorem 3.15), with the upper half false at fixed scale for the commute weighting (Proposition 3.16). The empirical flatness in m (§5.2) supports the rank form; the metric form reduces to one clean question in spectral geometry:

PROPOSITION 3.5 (The unnormalized differential does not collapse). *For nested multiplet-closed cutoffs $m < m'$ and every tangent v , $\|dF_{m'}(v)\|^2 = \|dF_m(v)\|^2 + \sum_{m < k \leq m'} |d\varphi_k(v)|^2 / \mu_k \geq \|dF_m(v)\|^2$, so $\sigma_{\min}(dF_m(x))$ is nondecreasing in m ; once F_{m_0} is an immersion (finite m_0 [5]), compactness gives $\inf_{m \geq m_0} \inf_x \sigma_{\min}(dF_m(x)) \geq c > 0$.*

The *unnormalized* lower bound is therefore automatic; the obstruction to two-sided uniformity lives entirely in the normalization—the angular lower bound $\|N_m(x) - N_m(y)\| \geq c d_M(x, y)$ must survive a denominator $\|F_m\| \rightarrow \infty$, which it does on flat tori (Proposition SM2.3) and is conjectured in general (Conjecture 3.4), while the *upper* constant diverges for commute (Proposition 3.16). What remains for the *metric* form of Conjecture 3.4 is the normalized lower bound beyond the torus, closed at fixed m by Theorem 3.10 and Lemma SM1.3. The *rank* form, however, is settled below the critical dimension: for $d \leq 3$ the commute angular ranking converges uniformly in m to the Green-kernel ranking (Theorem SM2.7), reducing Angular Preservation to positivity of a geometric Green-rank coefficient—equal to 1 on the circle and on two-point homogeneous spaces of dimension ≤ 3 (Corollary SM2.8).

A proof—or a counterexample forced by symmetry—is a self-contained problem we do not attempt; none of the paper’s proved results (Table 1.1) depends on its resolution.

3.1. Toward a proof of Conjecture 3.4. Proposition 3.3 controls truncation; the remaining question is whether row-normalization preserves the geometry the low modes carry. It does, and the mechanism is an exact identity. To set expectations plainly: the payoff of this subsection is a *transfer* theorem, not a proof of the conjecture—it reduces the conjecture, at fixed m , to two continuum hypotheses (local eigenmap bi-Lipschitzness, H1, and radius-flat transversality, H2), plus an injectivity hypothesis (H3) for the global form.

LEMMA 3.6 (Normalization preserves bi-Lipschitz geometry). *For nonzero $u, v \in \mathbb{R}^m$ with $\hat{u} = u/\|u\|$,*

$$\|\hat{u} - \hat{v}\|^2 = \frac{\|u - v\|^2 - (\|u\| - \|v\|)^2}{\|u\| \|v\|}.$$

If $F : (X, d) \rightarrow \mathbb{R}^m$ satisfies $L^{-1}d(x, y) \leq \|F(x) - F(y)\| \leq Ld(x, y)$ with $\rho^- \leq \|F\| \leq \rho^+$ and radius Λ -Lipschitz, then $\hat{F} = F/\|F\|$ obeys $\|\hat{F}(x) - \hat{F}(y)\| \leq (L/\rho^-)d(x, y)$, and if $\Lambda < 1/L$,

$$\|\hat{F}(x) - \hat{F}(y)\| \geq \frac{\sqrt{L^{-2} - \Lambda^2}}{\rho^+} d(x, y).$$

Proof. The identity combines $\|\hat{u} - \hat{v}\|^2 = 2 - 2\langle u, v \rangle / (\|u\| \|v\|)$ with $\langle u, v \rangle = \frac{1}{2}(\|u\|^2 + \|v\|^2 - \|u - v\|^2)$. The upper bound uses numerator $\leq \|u - v\|^2 \leq L^2 d^2$ and $\|u\| \|v\| \geq (\rho^-)^2$. For the lower bound, $(\|u\| - \|v\|)^2 \leq \Lambda^2 d^2$ and $\|u - v\|^2 \geq d^2 / L^2$ give numerator $\geq d^2 (L^{-2} - \Lambda^2)$; divide by $\|u\| \|v\| \leq (\rho^+)^2$. $\square$

The condition $\Lambda < 1/L$ is a quantitative radial degeneracy: the radius must vary slower than the embedding moves, and as $\Lambda \rightarrow 0$ (the limit of Corollary 3.1) normalization becomes a pure scaling with lower constant $1/(L\rho^+)$. On the 2-torus the identity is exact to 10^{-15} , but the margin is razor-thin ($\Lambda = 0.0918$ against lower stretch $A = 0.0920$), so the lemma's worst-case constant is near-vacuous there—while the pointwise tangential component never degenerates (0/9200 local pairs radially degenerate; measured local angular constant 2.2). This survives on a *non-homogeneous* manifold, where the radius genuinely varies and a radial fold could in principle occur: on the swiss-roll (a 2D manifold carried into $\mathbb{R}^{12}$, §4) only 1/29,481 local pairs across five draws are radially degenerate, the local angular bi-Lipschitz constant is finite (3.67 ± 0.40), and the radius Lipschitz constant sits well below the lower stretch ($\Lambda_{\text{med}}/A = 0.30 \pm 0.02$)—so (A2) holds off the homogeneous case too. Proposition 3.7 explains why, and Theorem 3.8 makes the resulting two-point tangential noncollapse (A2) the hypothesis itself, with $\Lambda < 1/L$ only one sufficient route to it—so the thin radius margin costs the theory nothing.

PROPOSITION 3.7 (Exact angular distortion). *Let F be differentiable with $r = \|F\| > 0$ and $\hat{F} = F/r$. Taking $y \rightarrow x$ in Lemma 3.6 gives, for every tangent vector v ,*

$$\|d\hat{F}(v)\|^2 = \frac{\|dF(v)\|^2 - dr(v)^2}{r^2} = \frac{\|dF(v) - \hat{u} dr(v)\|^2}{r^2}, \quad dr(v) = \langle dF(v), \hat{u} \rangle.$$

The angular stretch in a direction v is *exactly* the tangential part of $dF(v)$ —the component orthogonal to the radius—divided by r . So $\hat{F}$ is a local bi-Lipschitz immersion *iff* dF is nowhere purely radial ($dF(v) \nparallel F$ for $v \neq 0$), the lower stretch being $\inf_v \|dF(v) - \hat{u} dr(v)\| / (r\|v\|)$. This sharpens Lemma 3.6: the crude $\Lambda < 1/L$ merely forces the radial term $dr(v)$ below the total $\|dF(v)\|$, whereas the proposition shows what is really required is that the embedding's velocity never aligns with its position. Lemma 3.6 and Proposition 3.7 link the two halves of the principle and reduce the conjecture to standard ingredients.

The general transfer principle. Proposition 3.7 identifies the mechanism infinitesimally; its finite-distance form can be made the *hypothesis* itself. The result is a transfer theorem that knows nothing about manifolds, Laplacians, or eigenvectors—and of which the eigenmap theorem below (Theorem 3.10) is the commute-weighted special case. Any uniformly calibrated discrete embedding inherits local angular bi-Lipschitz geometry from a continuum embedding whenever the continuum displacement has a two-point tangential component bounded below; radius flatness is only one sufficient route to that tangential noncollapse.

THEOREM 3.8 (Deterministic angular transfer). *Let (X, d) be a metric space, $\{x_1, \dots, x_n\} \subset X$, and $F : X \rightarrow \mathbb{R}^m \setminus \{0\}$ with radius $R = \|F\|$ and angular map $N = F/R$. Let $Z_{n,i} \in \mathbb{R}^m \setminus \{0\}$ be any discrete embedding of x_i , with angular rows $\hat{Z}_{n,i} = Z_{n,i}/\|Z_{n,i}\|$, and fix a pair domain $\mathcal{P} \subset X \times X$ (e.g. $\{0 < d(x, y) \leq r_0\}$). Assume: (A1) $0 < \rho^- \leq R \leq \rho^+ < \infty$ on the relevant region; (A2) **two-point tangential noncollapse**: there are $0 < \tau \leq \beta < \infty$ with*

$$\tau d(x, y) \leq \left(\|F(x) - F(y)\|^2 - (R(x) - R(y))^2 \right)^{1/2} \leq \beta d(x, y) \quad \text{for all } (x, y) \in \mathcal{P}$$

(the middle quantity is exactly the part of the displacement that survives row-normalization; infinitesimally it is the tangential component of Proposition 3.7);
(A3) uniform calibration: there are a global scale $s_n > 0$, an orthogonal $Q_n \in O(m)$, and $E_n \rightarrow 0$ with $\max_{i \leq n} \|Q_n(s_n Z_{n,i}) - F(x_i)\| \leq E_n$. Then for every $\eta \in (0, 1)$, once $E_n < \rho^-/2$, every sampled pair in $\mathcal{P}$ with

$$d(x_i, x_j) \geq \theta_n(\eta) := \frac{4E_n \rho^+}{\eta \tau \rho^-} \longrightarrow 0$$

obeys

$$(1 - \eta) \frac{\tau}{\rho^+} d(x_i, x_j) \leq \|\hat{Z}_{n,i} - \hat{Z}_{n,j}\| \leq \left(\frac{\beta}{\rho^-} + \eta \frac{\tau}{\rho^+} \right) d(x_i, x_j).$$

Proof. The identity of Lemma 3.6 with $u = F(x)$, $v = F(y)$ reads $\|N(x) - N(y)\|^2 = [\|F(x) - F(y)\|^2 - (R(x) - R(y))^2] / (R(x)R(y))$, so (A1)–(A2) make N bi-Lipschitz on $\mathcal{P}$ with constants $a = \tau/\rho^+$ and $b = \beta/\rho^-$. Pairwise angular distances of $\hat{Z}$ are invariant under the scale s_n and the orthogonal Q_n , and by (A3) with normalization stability (Lemma SM1.2, using $\|F(x_i)\| \geq \rho^-$ in the denominator) each calibrated angular row lies within $2E_n/\rho^-$ of $N(x_i)$; hence each pairwise angular distance lies within $4E_n/\rho^-$ of the continuum one. For $d(x_i, x_j) \geq \theta_n(\eta)$ that error is at most $\eta a d(x_i, x_j)$; adding and subtracting it from the continuum bounds gives the claim. $\square$

COROLLARY 3.9 (Filtered spectral embeddings). *Let M and m (at a spectral gap) be as in Theorem 3.10, and let h be a positive Lipschitz spectral filter on the retained window ($H_m := \max_{k \leq m} h(\mu_k)$, L_h a Lipschitz constant). Define the filtered eigenmap $F_h(x) = (h(\mu_k) \varphi_k(x))_{k=1}^m$ and graph rows $Z_{n,i}^{(h)} = (h(\tilde{\lambda}_k) v_k(i))_{k=1}^m$ with $\tilde{\lambda}_k = \lambda_k/c_n$. Under the sup-norm spectral consistency of section SM1 (Lemma SM1.3 with $\text{diag}(\mu_k^{-1/2})$ replaced by $\text{diag}(h(\mu_k))$; same proof), hypothesis (A3) holds with high probability with*

$$E_{n,h} = H_m \epsilon_n + L_h \delta_n \sqrt{m} (\Phi_m + \epsilon_n) \longrightarrow 0.$$

Hence if F_h satisfies (A1)–(A2) on $\{d_M \leq r_0\}$, then with high probability the angular map of the h -filtered graph embedding is bi-Lipschitz to d_M , with the constants of Theorem 3.8, for all sampled pairs with $\theta_n(\eta) \leq d_M \leq r_0$. The commute filter $h(\lambda) = \lambda^{-1/2}$ recovers Theorem 3.10; diffusion $h(\lambda) = e^{-t\lambda}$ and plain $h \equiv 1$ are equally admissible at fixed m . The weighting ablation of §5.3—plain eigenmaps collapse as m grows while commute and diffusion stay flat—is therefore a statement about how each filter’s constants $(\tau, \beta, \rho^\pm)$ behave in m , not about the transfer mechanism, which is filter-independent.

THEOREM 3.10 (Conditional angular transfer for the commute-weighted eigenmap; local form of Conjecture 3.4). *Let M be a compact d -manifold whose graph Laplacian spectrally converges to M ’s ([11] for ε -graphs, [6] for k -NN graphs), and fix a mode-count m at a spectral gap of M . Assume **(H1)** the truncated eigenmap $F : x \mapsto (\varphi_k(x)/\sqrt{\mu_k})_{k=1}^m$ is L_0 -bi-Lipschitz onto its image on geodesic balls of radius r_0 —the quantitative truncated-eigenmap embedding property of [5, 2, 19]—and **(H2)** the associated truncated radius is Λ_0 -Lipschitz with $\Lambda_0 < 1/L_0$ and lies in $[\rho^-, \rho^+]$. Set*

$$A = \frac{\sqrt{L_0^{-2} - \Lambda_0^2}}{2\rho^+}, \quad B = \frac{2L_0}{\rho^-},$$

both independent of n . Then there is a resolution scale $\theta_n \rightarrow 0$ (explicitly, a constant multiple of the sup-norm spectral-consistency rate of [6]) such that, with high probability as $n \rightarrow \infty$, the graph angular map obeys $A d_M(x, y) \leq \|\hat{u}_x - \hat{u}_y\| \leq B d_M(x, y)$ for all sampled x, y with $\theta_n \leq d_M(x, y) \leq r_0$.

Proof (outline; complete proof in section SM1). This is the special case of Theorem 3.8 and Corollary 3.9 with $h(\lambda) = \lambda^{-1/2}$ and $\eta = 1/2$: (H1)–(H2) imply the tangential noncollapse (A2) with $\tau = \sqrt{L_0^{-2} - \Lambda_0^2}$ (positive because $\Lambda_0 < 1/L_0$) and $\beta = L_0$, giving $A = (1 - \eta)\tau/\rho^+$ and $B \geq \beta/\rho^- + \eta\tau/\rho^+$. Note (H2) is only one sufficient route: the theorem survives with (H2) replaced by (A2) directly, which is strictly weaker and is the condition the torus actually satisfies with a healthy margin (the discussion after Lemma 3.6). Self-contained proof: by (H1), (H2) and Lemma 3.6, the continuum angular map $N = F/\|F\|$ is bi-Lipschitz on r_0 -balls with constants $a = \sqrt{L_0^{-2} - \Lambda_0^2}/\rho^+$ and $b = L_0/\rho^-$ (Lemma SM1.1). The transfer to the graph needs sup-norm eigenvector convergence—the ∞ -norm rates of [6] rather than generic L^2 consistency—so that normalization survives where $\|F\|$ is smallest. It is applied to the random-walk eigenvectors, to which the graph angular map reduces *exactly* by per-node degree cancellation ($\Psi_i = \sqrt{k_i} \Psi_i^{\text{rw}}$; Step 0 (iv) of section SM1), so the transfer needs no degree-concentration hypothesis and holds under non-uniform sampling. After the graph-to-continuum eigenvalue scaling and a block-orthogonal alignment across the multiplets below the gap (both invisible to pairwise angular distances), the graph embedding is calibrated to F within an error $E_n \rightarrow 0$ uniformly over nodes (Lemma SM1.3); in particular the graph radius inherits $[\rho^- - E_n, \rho^+ + E_n]$, and the graph angle lands within $2E_n/\rho^-$ of N (Lemma SM1.2). The two-sided bound then follows with $A = a/2$ and $B = b + a/2 \leq 2L_0/\rho^-$ for every sampled pair above the scale $\theta_n = 8E_n/(a\rho^-)$. The cutoff is necessary, not an artifact: below the graph’s resolution, combinatorially twin nodes receive identical entries in every eigenvector below the localized twin-difference mode—in particular in every low mode the angular map uses—so $\hat{u}_i = \hat{u}_j$ while $d_M > 0$ (section SM1). $\square$

From local to global. Theorem 3.10 is local because (H1) is. One further hypothesis—the global sibling of the radial-fold transversality of Proposition 3.7—globalizes it.

COROLLARY 3.11 (Global conditional form). *Assume (H1)–(H2) and additionally (H3): the continuum angular map $N = F/\|F\|$ is injective on M . Let $c_0 := \min\{\|N(x) - N(y)\| : d_M(x, y) \geq r_0\}$, which is positive by (H3) and compactness. Then, with*

$$A_g = \min\left(A, \frac{c_0}{2 \text{diam } M}\right), \quad B_g = \max\left(B, \frac{2}{r_0}\right),$$

with high probability as $n \rightarrow \infty$ the graph angular map obeys $A_g d_M(x, y) \leq \|\hat{u}_x - \hat{u}_y\| \leq B_g d_M(x, y)$ for all sampled x, y with $d_M(x, y) \geq \theta_n$ (proof: section SM1, Step 5).

(H3) is genuinely extra: Portegies-style mode counts [19] make F itself injective, but $N(x) = N(y)$ requires only that F radially align two points ($F(x) = tF(y)$, $t > 0$), which injectivity of F does not exclude. Empirically it holds on the torus—the angular map folds nothing together (metric-evidence paragraph above). On the flat torus, in fact, nothing at all is left to assume:

COROLLARY 3.12 (Unconditional instance: the flat torus). *Let $M = \mathbb{R}^2/(2\pi\mathbb{Z})^2$ with $m = 4$ (the full first multiplet, a spectral gap). Then (H1)–(H3) hold, with*

explicit constants: the truncated eigenmap is the Clifford embedding $F(x, y) = \sqrt{2}(\cos x, \sin x, \cos y, \sin y)$, which is L_0 -bi-Lipschitz globally ($r_0 = \text{diam } M$) with $L_0 = \sqrt{2}$; the radius is constant, $R \equiv 2$, so $\Lambda_0 = 0$ and $\rho^\pm = 2$; and $N = F/2$ is injective. Consequently, with high probability as $n \rightarrow \infty$, the graph angular map of an ε -graph or k -NN sample of the flat torus is globally bi-Lipschitz to geodesic distance above the resolution scale, unconditionally, with $A = 1/(4\sqrt{2})$ and $B = \sqrt{2}$; the sharp continuum angular distortion is exactly $\pi/2$ (the chord-vs-arc ratio $2\sin(\delta/2)/\delta \in [2/\pi, 1]$ per coordinate—the source of the near-antipodal dip in the measured κ). The statement is basis-independent: any L^2 -orthonormal basis of the four-dimensional eigenspace is an $O(4)$ rotation of the Clifford one, invisible to angular distances (proof: section SM1, Step 6).

The measured graph-level $\kappa \approx 2.6$ (at $m=10$, which splits a multiplet; the gap-closed cutoffs on $\mathbb{T}^2$ are 4, 8, 12, 20, ...) sits above the $\pi/2 \approx 1.57$ continuum floor by finite- n and truncation slack; $m=4$ is the theoretically clean budget. The transfer theorem certifies only $B/A = 8$, so the measured $\kappa \approx 2.6$ sits between the sharp continuum $\pi/2$ and the certified 8—a gauge of how loose the transfer constants are.

COROLLARY 3.13 (Unconditional instance: the round sphere). *On $S^d \subset \mathbb{R}^{d+1}$ with $m = d+1$ (the $\ell=1$ eigenspace, eigenvalue d), $F \propto x$: the radius is constant and $N = F/\|F\| = x$ is the identity embedding, so N is globally bi-Lipschitz with the sharp constants $[2/\pi, 1]$ ($\|N(x) - N(y)\| = 2\sin(\theta/2)$, $\theta = d_{S^d}(x, y)$), unconditionally—a second constant-curvature instance ($\mathbb{T}^d$ likewise extends to $m = 2d$).*

More generally, on any compact homogeneous space transitivity makes the radius constant at every multiplet-closed cutoff, so (H2) holds automatically and the metric conjecture reduces there to (H1) and (H3) alone.

From metric to rank.. The conjecture’s currency is rank (Spearman); the theorems’ is distortion. The bridge is elementary and deterministic.

PROPOSITION 3.14 (Distortion controls rank). *Let $\mathcal{P}$ be a finite set of node pairs on which $A'd_M \leq \hat{D} \leq B'd_M$ holds for some embedded distance $\hat{D}$, write $\kappa = B'/A'$, and let $\hat{\mu}(\kappa)$ be the fraction of unordered pairs $\{P, Q\} \subset \mathcal{P}$ whose geodesic distances lie within ratio κ of each other ($\max/\min < \kappa$). Then, absent ties (ties, e.g. from integer hop distances, enter only through the standard tie corrections), Kendall’s $\tau(\hat{D}, d_M) \geq 1 - 2\hat{\mu}(\kappa)$ and Spearman’s $\rho_S \geq 1 - 3\hat{\mu}(\kappa)$ up to an $O(1/|\mathcal{P}|)$ finite-sample correction.*

Proof. A discordant pair-of-pairs ($d_P < d_Q$ but $\hat{D}_P > \hat{D}_Q$) forces $A'd_Q \leq \hat{D}_Q < \hat{D}_P \leq B'd_P$, i.e. $d_Q/d_P < \kappa$; so the discordant fraction is at most $\hat{\mu}(\kappa)$ and $\tau = 1 - 2(\text{discordant fraction}) \geq 1 - 2\hat{\mu}(\kappa)$. Daniels’ inequality $|3\tau - 2\rho_S| \leq 1$ [10] converts this to $\rho_S \geq (3\tau - 1)/2 \geq 1 - 3\hat{\mu}(\kappa)$. $\square$

Combined with Corollary 3.11, this reduces the rank clause of Conjecture 3.4, at fixed m , to (H1)–(H3) plus a distance-ratio condition on the substrate: $\rho_0 = 1 - 3\mu(\kappa) > 0$ whenever $\mu(\kappa) < 1/3$ (sub- θ_n pairs are a vanishing fraction and enter the bound as $o(1)$). The reduction is honest about its weakness: it is worst-case over all κ -distorted maps. For the small-ball distance law $f(r) \propto r^{d-1}$ on $[0, r_0]$ an exact computation (section SM1, Step 7) gives $\mu(\kappa) = 1 - \kappa^{-d}$, so the certificate $\tau \geq 2\kappa^{-d} - 1$ is non-vacuous only for $\kappa < 2^{1/d}$: a fixed distortion budget certifies exponentially less rank structure as the intrinsic dimension grows, and on substrates whose pairwise distances concentrate—the high- d regime—no finite distortion certifies any rank fidelity at all. This is a genuine ceiling for distortion-based guarantees, and

a second, rank-specific mechanism for the fidelity-vs-dimension decay of Paper I.b. That the measured decay is *linear* ($-0.157d$, Paper I.b), far milder than the worst case, says the eigenmap angle is far from the worst κ -distorted map: the empirical ≈ 0.9 levels remain stronger than what distortion alone implies, which is why that clause of the conjecture stays empirical.

Uniformity in the truncation: what is true, and what is false.. Everything so far is at fixed m . The uniformity-in- m question splits by filter, and both halves can now be answered in the continuum. Throughout, the cutoff is spectral and multiplet-closed: the map retains all eigenspaces with $\mu_k \leq \Lambda$, so $m = m(\Lambda)$, matching the spectral-gap convention.

THEOREM 3.15 (Uniform-in-truncation angular embedding: heat filter). *Let M be compact, smooth, without boundary, and for $t > 0$ let $F_{t,m}(x) = (e^{-t\mu_k/2}\varphi_k(x))_{k=1}^m$ be the heat-filtered eigenmap, with $N_{t,m} = F_{t,m}/\|F_{t,m}\|$. There is $t_0(M) > 0$ such that for every fixed $t \leq t_0$ there exist $m_0(t)$, $r_t > 0$, and $0 < a_t \leq b_t < \infty$ with*

$$a_t d_M(x, y) \leq \|N_{t,m}(x) - N_{t,m}(y)\| \leq b_t d_M(x, y) \quad \text{whenever } d_M(x, y) \leq r_t,$$

for every $m \geq m_0(t)$: the constants do not deteriorate as $m \rightarrow \infty$. (Proof: section [SM2](#), by heat-kernel diagonal asymptotics and C^2 tail control; the exponential filter is what makes the tail summable against the polynomial growth of eigenfunction norms.)

PROPOSITION 3.16 (Commute cutoff: tangential noncollapse, scale-dependent metric). *Let $F_\Lambda(x) = (\mu_k^{-1/2}\varphi_k(x))_{0 < \mu_k \leq \Lambda}$ be the commute-weighted spectral cutoff, $R_\Lambda = \|F_\Lambda\|$, $N_\Lambda = F_\Lambda/R_\Lambda$. There are $c > 0$ and Λ_0 such that for all $\Lambda \geq \Lambda_0$, uniformly over $x \in M$ and unit tangent v ,*

$$(\|dF_\Lambda(v)\|^2 - dR_\Lambda(v)^2)^{1/2} \geq c\Lambda^{d/4} :$$

the tangential component never collapses—it grows with the cutoff. Meanwhile $R_\Lambda \asymp \Lambda^{d/4-1/2}$ for $d > 2$ (and $\asymp \sqrt{\log \Lambda}$ for $d = 2$), so the angular derivative grows, $\|dN_\Lambda\| \gtrsim \Lambda^{1/2}$ ($d > 2$; $\sqrt{\Lambda/\log \Lambda}$, $d = 2$). Consequently no m -independent fixed-scale upper Lipschitz constant exists for the commute angular map: its natural resolution scale is the spectral wavelength $\Lambda^{-1/2}$. On S^1 this is exact: with all modes to frequency N , R_N is constant, $\|dF_N\|^2 = N$, and $\|dN_N\| \asymp \sqrt{N}$. (Proof: section [SM2](#), via the differentiated pointwise Weyl law of Lemma [SM2.1](#); the S^1 case needs no Weyl input and is exact.)

The two statements split the uniformity question cleanly. For the *heat/diffusion* filter—one of the two weightings the ablation of §5.3 finds flat in m —uniform-in-truncation local bi-Lipschitzness is a continuum theorem. For the *commute* filter the tangential mechanism survives uniformly, with room to spare, but the metric constants are intrinsically scale-dependent: high modes raise the local angular speed like $\sqrt{\Lambda}$, so the correct uniform metric object is bi-Lipschitzness *above the wavelength scale* $\Lambda^{-1/2}$ —equivalently, a mode budget matched to the scale of interest, which is the mode-budget phenomenon of Paper I.b handed a mechanism. The empirical “flat in m ” of §5.2 is thereby correctly read as *rank* stability under a fixed mode budget, not as an m -independent fixed-scale metric theorem. One gap remains at this point: the $\sqrt{\Lambda}$ angular-speed growth of Proposition 3.16 is filter-generic (the plain filter has the same order), so it explains the budget phenomenon but not the ablation’s dramatic *rank* separation between plain and commute (§5.3). The next results address that

gap: on a general manifold the plain filter *concentrates* (Proposition 3.17), and on the circle—the canonical homogeneous instance—its angular ranking provably collapses (Theorem 3.18).

PROPOSITION 3.17 (Plain-filter distance concentration). *Let $F_\Lambda^{(1)}(x) = (\varphi_k(x))_{0 < \mu_k \leq \Lambda}$ be the unweighted (plain-eigenmap) cutoff with angular map $N_\Lambda^{(1)}$, and fix $\delta > 0$. Then, uniformly over pairs with $d_M(x, y) \geq \delta$,*

$$\|N_\Lambda^{(1)}(x) - N_\Lambda^{(1)}(y)\|^2 = 2 - \frac{2E_\Lambda(x, y)}{\sqrt{E_\Lambda(x, x)E_\Lambda(y, y)}} = 2 + O(\Lambda^{-1/2}) :$$

all separated plain angular distances concentrate at the orthogonality value $\sqrt{2}$, and the vanishing correction is carried by the off-diagonal spectral function $E_\Lambda(x, y)$ —an oscillatory kernel with sign changes on the wavelength scale $\Lambda^{-1/2}$ and no monotone relation to d_M . (Proof: section SM2.)

Concentration alone does *not* imply that the ordering degrades: Spearman and Kendall correlations are invariant under a shrinking positive-affine rescaling, so a correction of the form $\varepsilon_\Lambda f(d_M)$ with $\varepsilon_\Lambda \rightarrow 0$ could preserve the rank order for every Λ . What the *oscillatory* form of E_Λ precludes is precisely that residual monotonicity. We make this exact on the canonical homogeneous instance, where E_Λ is the Dirichlet kernel and the collapse can be computed in closed form. For the *commute* filter the same expansion instead gives correction kernel $G_\Lambda(x, y) = \sum_{0 < \mu_k \leq \Lambda} \mu_k^{-1} \varphi_k(x) \varphi_k(y)$, low-frequency dominated (for $d \leq 3$ it converges off the diagonal to the Green’s function)—a coherent, distance-structured kernel; on $\mathbb{S}^1$ it is the monotone periodic Green’s function, so the commute ordering there is preserved.

THEOREM 3.18 (Exact plain-filter rank collapse on $\mathbb{S}^1$). *On the circle $\mathbb{S}^1 = \mathbb{R}/2\pi\mathbb{Z}$, let the plain eigenmap retain the first N complete Fourier multiplets, $F_N(x) = (\cos x, \sin x, \dots, \cos Nx, \sin Nx) \in \mathbb{R}^{2N}$, with angular map $\hat{F}_N = F_N/\|F_N\|$ (so $\|F_N\|^2 \equiv N$). Let X, Y be independent uniform points, $\Theta = d_{\mathbb{S}^1}(X, Y) \sim \text{Unif}[0, \pi]$, and $Q_N(\Theta) = \|\hat{F}_N(X) - \hat{F}_N(Y)\|^2$. Then the population Spearman and Kendall correlations against geodesic distance both vanish:*

$$\rho_S(\Theta, Q_N(\Theta)) \xrightarrow{N \rightarrow \infty} 0, \quad \tau_K(\Theta, Q_N(\Theta)) \xrightarrow{N \rightarrow \infty} 0.$$

Thus on $\mathbb{S}^1$ the plain angular ordering retains asymptotically no geodesic information, even though every separated distance concentrates at $\sqrt{2}$ (Proposition 3.17). (Proof: section SM2.)

The mechanism in one line: the $1/\mu$ commute weighting is a low-pass filter on the *correction* kernel that carries the ranks, while equal weighting hands the correction to the oscillatory top of the spectral window—which is why the ablation separates the filters at the rank level (0.06 vs 0.89 at $m=20$, §5.3) even though their crude metric orders coincide. For graph transfer at growing $m = m(n)$, the calibration of Lemma SM1.3 extends whenever $\mathcal{E}_n(\mu_m) \gamma_m^{-1} \rightarrow 0$ (γ_m the gap isolating the retained block; $\mathcal{E}_n$ the graph-consistency error, polynomially growing in μ_m), and a $C^{0,1}$ eigenvector theory [7] is the route to shrink the resolution cutoff toward the connectivity scale (supplement, section SM2).

What remains is delimited. (H1) is subtle: Jones–Maggioni–Schul [16] guarantee bi-Lipschitz charts only from a well-chosen d -subset of eigenfunctions per ball, and low-spectrum multiplicity makes a truncation inside a multiplet basis-dependent—so

m is taken at a spectral gap (the $m=4$ of Corollary 3.12) or the map phrased through spectral projectors. **(H2)** is, by Proposition 3.7, sharper than radius flatness: the eigenmap’s velocity must be nowhere purely radial, with (A2) its two-point form and the hypothesis to verify in practice; it holds on the tested manifolds, homogeneous and not (§3), and whether the eigenmap image ever develops a radial fold on some manifold is the sharp remaining question. **(H3)** is genuinely extra: injectivity of F [19] does not exclude a radial alignment $F(x) = tF(y)$, so the global statement is carried by (H3) (Corollary 3.11)—while on the torus (H1) holds globally and Corollary 3.12 needs none of the chart hypotheses (H1)–(H3) (the sampling model and spectral consistency of Theorem 3.10 are still assumed).

Status. Table 1.1 records what is proved, conditional, empirical, and open; at fixed m the rank clause reduces to (H1)–(H3) plus a distance-ratio condition on the substrate (Corollary 3.11, Proposition 3.14), and uniformity in m is settled in the continuum by filter. The honest gap: Corollary 3.1 explains why the full unnormalized radius *dies* ($d \geq 3$), not why the angle *lives*—angular fidelity persists ($\rho \approx 0.88$ – 0.90) even on quasi-1D graphs where the corollary does not apply, so the angular half is carried by the eigenmap-embedding mechanism. That mechanism is analyzable: if the low eigenvectors converge to Laplace–Beltrami eigenfunctions [4, 11] and eigenfunction coordinates furnish locally bi-Lipschitz charts [16], geodesic fidelity should follow from the chart geometry—the natural route to Conjecture 3.4, whose residual difficulty is isolated to the normalization by Proposition 3.5. Together these are the **Keep-the-Angle Principle**: keep $\hat{u}_i$, discard r_i —and, under strongly non-uniform sampling, keep the $\alpha=1$ density-normalized angle (Paper I.b).

Dimension gating. Corollary 3.1 is a $d \geq 3$ theorem; $d = 2$ is the recurrent borderline, where the degeneracy appears with logarithmic corrections and we treat it empirically (§5.2). In (quasi-)1D, resistance equals path length, so the radius *stays* geodesic-informative and Corollary 3.1 correctly does not apply—a prediction, not a failure, confirmed in §5.2.

TABLE 4.1

Protocol by section. n : nodes; m : retained nodes; n_a : anchors; D : independent graph draws reported (mean $\pm$ sd when $D > 1$; single-draw rows marked $\dagger$ per the headline-uncertainty policy below).

Section	substrate(s)	graph	n	m	n_a	D
§5 core	2-torus	kNN	2000	5–80	150	$1^\dagger$ (5 at $m=10$)
§5.2	2-torus	kNN	2000–4000	5–80	200	5 ($m=10$)
§5.3	2-, 3-torus	kNN	2000/3000	10–40	150	$1^\dagger$ (10 ctrl)

4. Methods. Where a manifold ground truth exists we report both the measured consensus dimension d_G and the true d_M (2-torus $d_M=2$; 3-torus 3; swiss-roll 2; king-lattice 2). On the controls the estimators read a few percent low (a known finite-sample curvature bias): the consensus $d_G \approx 1.94$ on the 2-torus, with the individual estimators 5–8% below d_M at $n=2000$. All dimension-indexed claims use d_G consistently: estimator bias shifts the axis, not the ordering of the Paper I.b trend.

Estimators. Ball-growth $N(r) \sim r^d$; spectral dimension from the heat-kernel return probability $P(t) = \langle e^{-Lt} \rangle \sim t^{-d_s/2}$; effective rank via local-PCA participation ratio. Low inter-estimator spread $\Rightarrow$ clean manifold, and the agreed value is our measured intrinsic dimension d . **Angular metric.** Spearman ρ between embedded pairwise distances and true graph geodesics for: *angle* (chordal distances

between unit-normalized low-mode embeddings, §3), *random* (m random modes; control), *magnitude* (radius only), and *Isomap* (classical MDS of the graph geodesics, an oracle given the geodesics). **Uncertainty.** Each ρ is a rank correlation over $\binom{n_{\text{anchor}}}{2}$ node pairs ($n_{\text{anchor}} = 150$ for the substrate benchmarks, 200 for the torus metric tests (§3.1)—19,900 pairs). Bootstrap 95% confidence intervals over pair resampling are tight ($\leq \pm 0.005$ on the headline numbers), but pairs share anchor nodes and are not independent, so we treat pair-resampling CIs as optimistic floors; the dominant source of variation is the graph draw itself, so for headline ρ we also report spread over independent draws (2-torus, $m=10$: 0.914 ± 0.009 over five draws). The reference torus therefore appears at slightly different angle- ρ across tables (0.89–0.93) as mode-count, anchor count, and graph draw vary by section; each table uses its section’s protocol ($m=20$, $n_{\text{anchor}}=150$ unless stated). **Substrates.** Arity-2/3 hypergraph rewriting (clique-expanded), plus tori, king-lattices, Poisson causal sets, and swiss-roll embedding samples. The swiss-roll case is a synthetic 2D manifold—intrinsic coordinates (u, v) , u the roll angle and v the height—carried by a random orthogonal map into $\mathbb{R}^{12}$ with isotropic noise ($\sigma=0.03$) and connected by a $k=10$ nearest-neighbour graph on $n=2000$ ambient points. It stands in for sequential/embedding-derived data (each item a point on a low- d semantic manifold, consecutive items nearby), not a specific text corpus. The Lorentzian causal set is an $n=2000$ Poisson sprinkling in a 1+1D Minkowski diamond (light-cone coordinates u, v uniform on the unit square); $p \prec q$ iff $u_p < u_q$ and $v_p < v_q$, and each point is linked to its $k=6$ nearest causal-past neighbours in proper time $\tau^2 = \Delta t^2 - \Delta x^2$, symmetrized for the undirected geodesic BFS—so edges encode short causal *relations* (nearest past neighbours in proper time, which need not be Hasse links), and the intrinsic $d=2$ is the sprinkling dimension.

5. Results.

5.1. The core result.

Estimators recover ground-truth dimension within ≈ 0.3 . On a 2-torus the polar decomposition of the low-mode embedding (Fig. 5.1) gives:

m	full (mag. $\times$ dir.)	angle only	magnitude only
4	0.820	0.928	0.024
8	0.703	0.933	0.027
12	0.595	0.900	0.030
20	0.525	0.891	0.019

The cutoffs are multiplet-closed (the gap-closed budgets 4, 8, 12, 20 on $\mathbb{T}^2$), and each entry is a mean over five independent graph draws (angle standard deviation ≤ 0.009 ; e.g. $m=20$ gives 0.891 ± 0.009). The angle carries the geometry (≈ 0.9 , flat across the admissible schedule); the magnitude carries little geodesic information in these benchmarks; the full embedding is worse and decays. A random-mode basis reconstructs geodesics at $\rho \approx 0$ (Fig. 5.2); averaged over ten random draws $\rho = 0.08 \pm 0.16$, individual draws occasionally catching a low mode and spiking to ~ 0.4 .

5.2. Verifying Corollary 3.1 and Conjecture 3.4.

We check the two claims directly. The angle- ρ is flat in graph size while the full commute distance decays, and the truncated radius tracks degree ($\rho(\|\Psi_i\|, 1/\sqrt{k_i}) = 0.79\text{--}0.87$ across $m = 5\text{--}40$; radius-only $\rho \leq 0.08$). For the full *unnormalized* commute radius the degeneracy is exact, Corollary 3.1 holding cleanly at $d = 3$ ($\rho(R, 1/k_i + 1/k_j) \rightarrow 0.98$); $d = 2$ is borderline (log corrections); the quasi-1D case correctly shows no degeneracy, as the dimension gating predicts. Conjecture 3.4: in an m -sweep at $n = 4000$, angle- ρ is flat ≈ 0.93 across $m = 5\text{--}80$ while the full embedding decays $0.773 \rightarrow 0.437$.

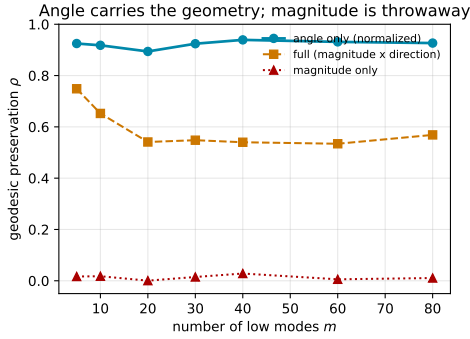


FIG. 5.1. Spearman correlation with graph geodesics on a 2-torus ($n = 2000$) versus mode-count m . Angle carries the geometry; magnitude carries little geodesic signal; the full commute embedding decays with m .

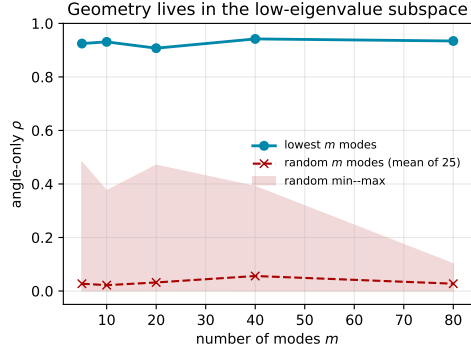


FIG. 5.2. Geometry lives in the low-eigenvalue subspace: lowest- m modes preserve geodesics; a random-mode basis of equal size does not. The random curve is the mean over 25 draws, the band its full min-max envelope—even the best-case random draw stays far below the low-mode curve.

5.3. Which weighting carries the angle?. The direction depends on the per-mode weight, so we ablate it on the torus ($d=2$) and a 3-torus ($d=3$). Plain eigenmaps ($w_k=1$) give an angle that *collapses* with mode count—angle- ρ falls from 0.554 ± 0.013 at $m=10$ to 0.081 ± 0.012 at $m=20$ (five independent draws, $d=2$)—because equal weighting lets high modes swamp the direction. The commute weight $1/\sqrt{\lambda_k}$ is flat instead ($0.918 \pm 0.006 \rightarrow 0.896 \pm 0.006$), as is diffusion $e^{-\lambda_k t}$; both suppress high modes, which is what the geometry needs. The “flat in m ” property is thus a property of the weighting, not of angles in general—and it selects the commute (or diffusion) weight, not the plain eigenmap. Proposition 3.17 now supplies the continuum mechanism: plain angular distances concentrate at $\sqrt{2}$ with only an oscillatory, geometry-blind correction, while the commute correction is the (low-pass, Green-type) kernel that carries the ranks. Degree correction, by contrast, is *invisible* to the angle: dividing or multiplying each row by $\sqrt{k_i}$ (the diffusion-map convention and its opposite) leaves angle- ρ unchanged to three digits (0.889 either way), because a per-node radial factor cancels under row-normalization—a direct confirmation that the degree lives in the radius (Corollary 3.1), not the angle.

6. Discussion and Conclusion. The primary result is basis-level and substrate-agnostic. The angular coordinate of the low-Laplacian embedding preserves geodesics while the radial coordinate is degree/density nuisance—von Luxburg-degenerate for the full, unnormalized commute radius at $d \geq 3$ (Corollary 3.1), and empirically degree-tracking for the truncated, normalized radius actually in use (§5.2). This gives Ng–Jordan–Weiss row-normalization a geodesic justification complementary to the cluster-separation theory of [23], and places it in the same scale-from-direction decomposition that recurs in vector and KV-cache compression [14]—the quotient, not the discard, being what recurs. The empirical reach of the basis across substrate families, its role as a manifold diagnostic, the compression scope boundary (angular quantization fails for attention keys, where scale is signal, not nuisance—condition (A2) of Theorem 3.8), and a scoped physical interpretation are developed in the companion Paper I.b [1].

Open problems.. Proving Conjecture 3.4 (rank form, and the bi-Lipschitz strengthening) is now reduced, at fixed m , to the chart hypotheses (H1)–(H3) plus a distance-ratio condition for rank (Corollary 3.11, Proposition 3.14); the open core is the *scale-dependent* commute uniformity—fixed-scale uniformity in m is proven for the heat filter and refuted for the commute filter (Theorem 3.15, Proposition 3.16)—and graph transfer at growing spectral windows (section SM2). Beyond the flat torus, where Corollary 3.12 settles them, (H1)–(H3) must be checked directly—the obstruction being the normalized upper bound, not the differential lower bound, which never collapses (Proposition 3.5). Also open: extending Corollary 3.1 to the truncated, normalized radius used in practice; synthesizing clean manifolds above $d \approx 2$; a dimension-adaptive basis that tunes mode-count and diffusion time against the fidelity decay; and whether the diagnostic transfers to learned representations (a preliminary probe is in the repository).

Reproducibility.. The confirming experiments of this paper—the torus core result and controls, the theorem verification, the weighting ablation, and the truncation/normalization checks—are fully reproducible locally (NumPy/SciPy/NetworkX, CPU), each script emitting the exact committed result JSON in the accompanying repository, <https://github.com/ahb-sjsu/the-angular-observer> (pre-registered as PREREG_RUNG3/4 with dated amendments). The extended empirical program and its reproducibility tiers are documented in the companion Paper I.b [1].

Appendix A. Proof of Corollary 3.1.

The diagonal statement $L_{ii}^+ \rightarrow 1/k_i$ does not follow from the pairwise resistance limits alone: $R(i, j) = L_{ii}^+ + L_{jj}^+ - 2L_{ij}^+$ fixes only those combinations, so a summation argument must control the error it accumulates. Two inputs are used, both explicit in the corollary’s statement: (i) the *uniform multiplicative* form of the von Luxburg–Radl–Hein degeneracy—for $d \geq 3$ graphs in their model there is a sequence $\eta_n \rightarrow 0$ with

$$R(i, j) = \left(\frac{1}{k_i} + \frac{1}{k_j} \right) (1 + \eta_{ij}), \quad \sup_{i \neq j} |\eta_{ij}| \leq \eta_n$$

(this multiplicative-uniform form is not stated verbatim by [26] (their Corollaries 8–9 and Propositions 29–30): they prove an *additive* deviation bound on the rescaled commute distance, uniform over $i \neq j$, together with two-sided degree bounds; dividing the additive bound by $1/k_i + 1/k_j$ and using the degree bounds yields the displayed η_n); and (ii) a bounded degree ratio, $k_{\max}/k_{\min} \leq K$ for all n (their graph models satisfy this with high probability). Write $S := \sum_j 1/k_j$ and note $S/n \leq 1/k_{\min} \leq K/k_i$ for every i by (ii).

Step 1 (row sums). L^+ annihilates constants ($L\mathbf{1} = 0$ and L^+ shares the range of L on $\mathbf{1}^\perp$), so $\sum_j L_{ij}^+ = 0$: the gauge. Summing $R(i, j)$ over $j \neq i$ and using the gauge,

$$\sum_{j \neq i} R(i, j) = (n-1)L_{ii}^+ + (\text{tr } L^+ - L_{ii}^+) - 2 \sum_{j \neq i} L_{ij}^+ = nL_{ii}^+ + \text{tr } L^+,$$

since $\sum_{j \neq i} L_{ij}^+ = -L_{ii}^+$. By (i) the same sum equals

$$\sum_{j \neq i} \left(\frac{1}{k_i} + \frac{1}{k_j} \right) (1 + \eta_{ij}) = \frac{n-2}{k_i} + S + e_i, \quad |e_i| \leq \eta_n \left(\frac{n}{k_i} + S \right).$$

Equating and dividing by n ,

$$(A.1) \quad L_{ii}^+ = \frac{1}{k_i} + C_n - \frac{2}{nk_i} + \frac{e_i}{n}, \quad C_n := \frac{S - \text{tr } L^+}{n} \quad (\text{independent of } i).$$

Step 2 (trace identity kills C_n). Summing (A.1) over i : $\text{tr } L^+ = S + nC_n - \frac{2}{n}S + \frac{1}{n} \sum_i e_i$. Substituting $nC_n = S - \text{tr } L^+$ and solving,

$$\text{tr } L^+ = S \left(1 - \frac{1}{n}\right) + \frac{1}{2n} \sum_i e_i, \quad \text{hence} \quad C_n = \frac{S}{n^2} - \frac{1}{2n^2} \sum_i e_i.$$

Since $|\sum_i e_i| \leq \eta_n \sum_i (n/k_i + S) = 2\eta_n nS$, we get $|C_n| \leq S/n^2 + \eta_n S/n \leq (1/n + \eta_n) K/k_i$ for every i , using (ii).

Step 3 (conclusion). Feeding the bounds on C_n and e_i/n back into (A.1),

$$\left| L_{ii}^+ - \frac{1}{k_i} \right| \leq \frac{(1/n + \eta_n)K}{k_i} + \frac{2}{nk_i} + \frac{\eta_n}{k_i} (1 + K) \leq \frac{C(K) (\eta_n + 1/n)}{k_i},$$

so $k_i L_{ii}^+ \rightarrow 1$ uniformly over nodes, i.e. $L_{ii}^+ = (1/k_i)(1 + o(1))$ and $\|\Phi_i\| = \sqrt{L_{ii}^+} = (1/\sqrt{k_i})(1 + o(1))$.

Without (ii) the conclusion weakens gracefully: the same computation gives $L_{ii}^+ = 1/k_i + O((\eta_n + 1/n)S/n)$, an *additive* error at the harmonic-mean-degree scale, so the degeneracy survives at the pairwise level ($R \approx 1/k_i + 1/k_j$, which is (i)) even when per-node relative control is lost. The $j=i$ term ($R(i, i) = 0$ while $1/k_i + 1/k_j = 2/k_i$) is excluded from the sums, which is why the $(n-2)/k_i$ bookkeeping appears.

Supplementary materials. An online supplement collects the longer proofs and the extended continuum theory: the conditional transfer theorem (§SM1); the truncation-uniformity results, the uniform lower angular bound and exact upper divergence on flat tori, the spectral-filter phase diagram, and the Green-kernel rank-limit theorem (§SM2). Every statement in the main article is self-contained; each supplementary proof is cross-referenced from the result it establishes.

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